Sky Maps and Catalogues

MAIS Physics & Astronomy Club (PAC) · Olympiad lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 2 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to astronomical measurements. Two vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice.

▼ Contents & learning path

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1. A sky map is a model of the celestial sphere

The sky appears as a sphere surrounding the observer. Right ascension (RA) and declination (Dec) label positions on that sphere. RA is measured eastward along the celestial equator in hours, minutes and seconds; 24 h = 360°, so 1 h = 15°. Dec is measured north or south of the equator in degrees.

A chart projects the sphere onto a plane. Every projection distorts something: area, shape, distance or direction. Near the celestial poles, RA lines converge, so a fixed RA interval corresponds to a smaller sky distance.

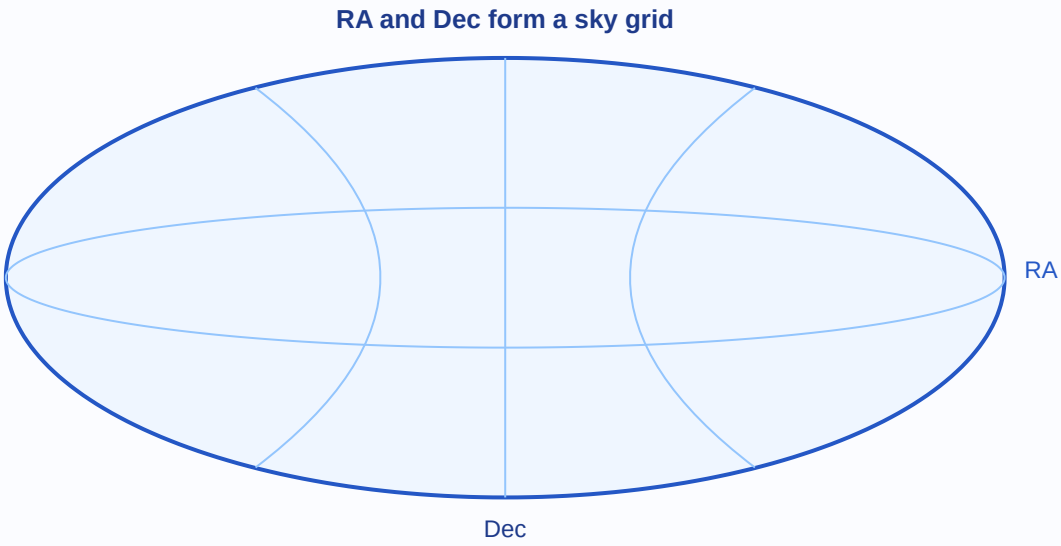


Figure 1. Charts project this grid; the projection distorts shapes and areas.

2. Catalogues are databases, not pictures

A catalogue records standardized measurements: identifier, position, epoch, magnitude, band, proper motion, parallax, spectral type and uncertainties. A chart is only a visualization. When using a catalogue, check the proper-motion epoch (when the position was measured), the reference-frame equinox (the coordinate grid orientation), the photometric band, and the uncertainty. A position from 1950 is not the same as a position from 2000 if the star has proper motion, even when both are expressed in the same equinox.

Common identifiers include HD, HIP, Gaia, Messier and NGC/IC. Cross-identification can be nontrivial: a close pair, a variable star or a crowded field may have multiple entries.

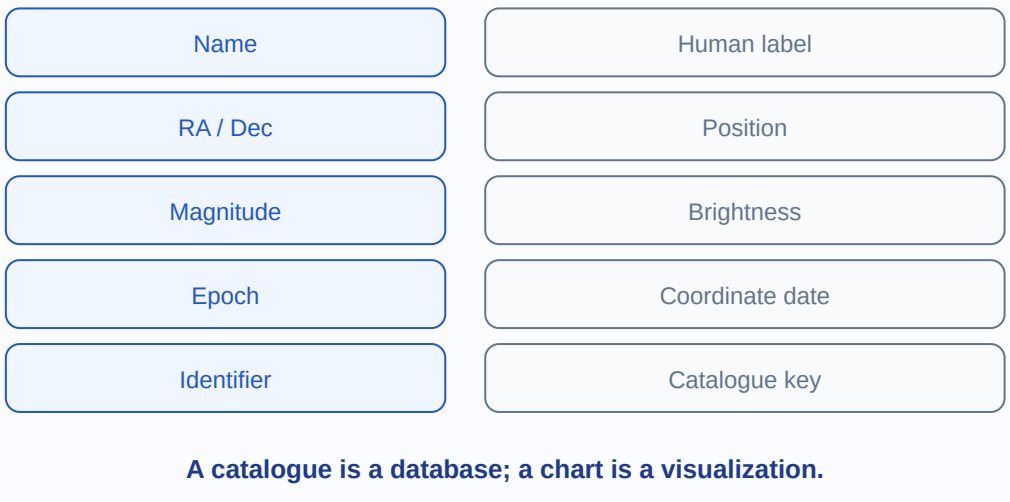


Figure 2. Always check epoch, equinox, band and uncertainty when comparing catalogue entries.

3. Reading a chart

Identify the projection, orientation, scale and magnitude limit. North and east may not be up and left. The magnitude limit controls how many stars are shown; a deeper chart is not automatically better for finding a target because it can be crowded. Use bright asterisms as anchors, then star-hop to the target.

For a finder chart, mark the target, comparison stars and field size. State the epoch and the source of the positions.

4. Proper motion and epoch

Stars move. Proper motion μ in arcsec/year changes position over time:

$$\Delta\theta = \mu\,\Delta t. \tag{1}$$

For high-precision work, precession, nutation, parallax and radial motion also matter. For visual star-hopping, a recent catalogue and a chart epoch within a few decades are usually sufficient, but high-proper-motion stars can shift noticeably.

5. Magnitude limits and selection effects

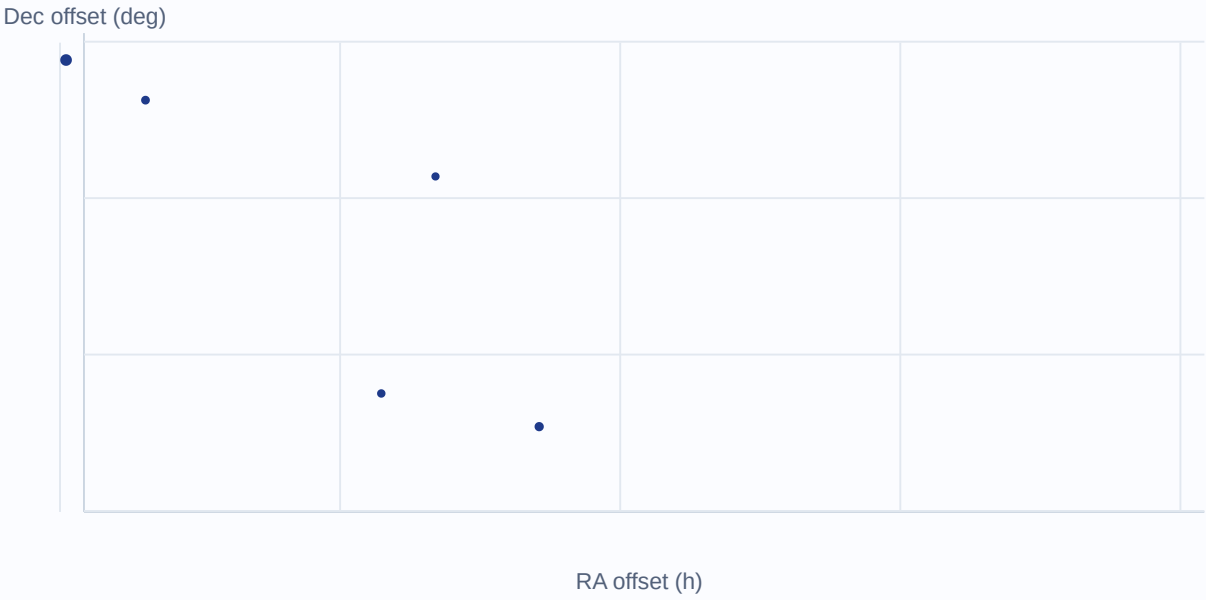
A magnitude-limited chart includes all stars brighter than a limit, regardless of distance. This is not a volume-limited sample. Intrinsically bright stars can be seen from far away; faint stars are included only nearby. This selection effect matters when interpreting star counts or comparing clusters.

INTERACTIVE EXPERIMENT

Filter a catalogue and read a chart.

A small synthetic catalogue is plotted in equatorial coordinates. Marker size encodes brightness; the grid spacing is 2 h in RA and 20° in Dec.

Stars in chart	Brightest in chart	Field width
5	**1.9**	**8 h × 60°**



Catalogues list positions and magnitudes; charts project them. A magnitude-limited chart is not the same as a volume-limited sample.

Try this. Raise the magnitude limit. Why does the number of stars grow quickly? Move the chart centre to a sparse region.

Snapshot of the experiment. Open the HTML edition to change the controls.

6. From catalogue to observation

A good workflow is: identify the target in a catalogue, check epoch and band, plot a finder chart with known orientation, choose comparison stars, and record the source. If the target is not where expected, check proper motion, coordinate units, RA/Dec order, and whether the chart is mirrored.

A reliable method. Always ask: What epoch? What band? What projection? What magnitude limit? What is the uncertainty? A chart without metadata is a guess.

7. Lecture + Problems: predict, calculate, check

Ten original problems: **[1]** foundation, **[2]** multi-step Olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. Explain assumptions and keep units. Every problem has a worked solution.

[1] Problem 1 — RA to degrees. Convert RA 6 h to degrees. Convert 90° to hours of RA.

Worked solution →

[1] Problem 2 — Catalogue fields. List four fields you should check before using a catalogue position for a finder chart.

Worked solution →

[2] Problem 3 — Proper motion. A star has proper motion 0.5″/yr. How far does it move in 50 years? Is this important for a 1′ finder field?

Worked solution →

[2] Problem 4 — Chart scale. A chart has scale 1 cm = 1°. A target is 3.5 cm from a reference star. What is the angular separation? If the chart is mirrored, what changes?

Worked solution →

[2] Problem 5 — Magnitude limit. A chart shows stars to magnitude 6. Another shows stars to magnitude 8. Using $N \propto 10^{0.6m}$, estimate the ratio of star counts.

Worked solution →

[1] Problem 6 — Epoch. Why is a catalogue labelled J2000 not automatically correct for today's pointing? Name two effects.

Worked solution →

[2] Problem 7 — RA separation. Two stars at the same declination differ by 2 minutes of RA. Using the small-angle approximation, what is their angular separation at declination 0°? What is it at declination 60°?

Worked solution →

[2] Problem 8 — Cross-identification. A target has two catalogue entries 2″ apart with different magnitudes. Give two possible explanations and one way to decide which is correct.

Worked solution →

[3] Problem 9 — Finder-chart design. Design a finder chart for a magnitude 9 variable star in a crowded field. State the field size, magnitude limit, orientation and comparison-star strategy.

Worked solution →

[3] Problem 10 — Selection effect. Why does a magnitude-limited star count not directly give the spatial density of stars? What would a volume-limited sample require?

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IOAA syllabus. Official problems and constants are available through the IOAA past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.